# Cloud Computing TASK – 1

1. **Sign up for a free AWS or Azure account and log in to the Management Console. Take a screenshot of your dashboard showing your name or email in the top bar.**

   **Ans :**

   - Open a web browser and visit https://aws.amazon.com/.
   - Click Create a Free Account.
   - Enter your email address, choose an AWS account name, and create a password.
   - Verify your email address using the verification code sent to your inbox.
   - Provide the required contact information and billing details to complete the registration process.
   - Complete the identity verification process by entering the OTP sent to your mobile phone.
   - Select the Basic Support Plan (Free).
   - After the account is activated, click Sign In to the Console.
   - Log in using your registered email address and password.
   - The AWS Management Console dashboard opens successfully.
   - Confirm that your name or email address is visible in the top-right corner of the dashboard.

2. **In the AWS or Azure Management Console, locate and list 5 key services (such as Compute, Storage, Database, Networking, Analytics). For each, write one line describing what it does and name one real app (like Instagram, Zomato, Flipkart) that could use it.**

   **Ans :**

| Service | Description | Example Application |
|---|---|---|
| Amazon EC2 (Compute) | Provides virtual servers in the cloud to run applications and websites. | Instagram uses compute servers to run its web and mobile application backend. |
| Amazon S3 (Storage) | Stores files, images, videos, backups, and other data securely in the cloud. | Flipkart stores product images, documents, and media files. |
| Amazon RDS (Database) | Provides managed relational databases for storing and managing application data. | Zomato stores user accounts, restaurant information, and order details. |
| Amazon VPC (Networking) | Creates a secure private network for cloud resources and controls network access. | Netflix uses secure networking to connect its cloud resources safely. |
| Amazon Athena (Analytics) | Analyzes data stored in Amazon S3 using SQL without managing servers. | Amazon analyzes customer and sales data to generate business insights and reports. |

3. **Use the search bar in your AWS or Azure Console to find the S3 (AWS) or Blob Storage (Azure) service. Write down the steps you took to locate it and describe in 2-3 lines how a music app like Spotify could use this service to store user-uploaded files.**

   **Ans :**

   - Log in to the AWS Management Console.
   - Locate the Search bar at the top of the console.
   - Type S3 in the search bar.
   - Select Amazon S3 from the search results.

- The Amazon S3 dashboard opens, where you can create and manage storage buckets.

Spotify could use Amazon S3 to securely store user-uploaded audio files, profile pictures, playlists, and other media. S3 provides scalable and durable cloud storage, allowing users to upload and access their files quickly from anywhere while ensuring the data remains safe and highly available.

4. **Pick one key characteristic of cloud computing (on-demand self-service, broad network access, resource pooling, rapid elasticity, or measured service) and explain, in your own words, how it benefits users of a popular app you use daily (e.g., Paytm, BookMyShow, IRCTC).<br><br><em><strong>Hint:</strong> Think about how these apps handle sudden spikes in users or data.</em>**

   **Ans :**

   - Key Characteristic: Rapid Elasticity
   - Rapid elasticity means cloud resources can automatically increase or decrease based on the number of users. This helps apps handle sudden increases in traffic without slowing down or crashing.
   - For example, BookMyShow experiences a large number of users when tickets for a popular movie or live event go on sale. Cloud computing automatically provides additional servers to manage the extra traffic, allowing users to book tickets smoothly. Once the demand decreases, the extra resources are released, which helps reduce costs.